Aman Dimri

Graduate Researcher, Stony Brook University

ATLAS Collaboration, CERN • STAR Collaboration, Brookhaven National Laboratory

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Research Interests

Experimental heavy-ion physics; quark–gluon plasma; collective phenomena and relativistic hydrodynamics; radial and anisotropic flow; event-by-event fluctuations; nuclear structure in relativistic collisions; initial-state geometry and QGP transport properties.

Education

Ph.D. in Nuclear Chemistry 2021–Present

Stony Brook University, Stony Brook, NY, USA
Department of Chemistry
Advisor: Prof. Jiangyong Jia

Integrated M.Sc. in Physics (Five-Year Combined Bachelor's–Master's Program) 2016–2021

National Institute of Science Education and Research (NISER), Bhubaneswar, India
Minor in Computer Science
Thesis: *Particle Production in High-Energy Collisions*
Advisor: Prof. Bedangadas Mohanty

Research Experience

Graduate Researcher — Heavy-Ion Physics 2021–Present

Stony Brook University; ATLAS Collaboration, CERN; STAR Collaboration, BNL

- Developing an ATLAS analysis of event-by-event mean transverse-momentum fluctuations in Pb+Pb collisions, aimed at disentangling intrinsic and geometry-driven contributions across collision centrality.
- Served as the main analyzer for multi-particle cumulant measurements in O+O and Ne+Ne collisions within the ATLAS primary analysis group.
- Performed data analysis for the ATLAS measurement of differential radial flow in Pb+Pb collisions as a contributing member of the primary analysis group.
- Proposed the constant-offset interpretation of $v_0(p_T)$ induced by global multiplicity fluctuations and derived the underlying analytical framework.
- Investigated the effects of nuclear quadrupole deformation and triaxiality fluctuations on initial-state geometry using Monte Carlo simulations.

Master's Research — Particle Production in High-Energy Collisions 2020–2021

National Institute of Science Education and Research (NISER), Bhubaneswar, India
Studied particle production in relativistic heavy-ion collisions using the Hadron Resonance Gas model together with transport and hydrodynamic calculations.

Publications

1. **The shape of differential radial flow $v_0(p_T)$, not its zero-crossing, carries physical information** [*NST*] [arXiv]
   S. Bhatta, **A. Dimri**, and J. Jia. *Nucl. Sci. Tech.* **37**, 163 (2026).

2. **Measurement of the azimuthal anisotropy of charged particles in $\sqrt{s_{NN}}$ = 5.36 TeV ^{16}O+^{16}O and ^{20}Ne+^{20}Ne collisions with the ATLAS detector** [*PRC*] [arXiv]
   ATLAS Collaboration. *Phys. Rev. C* **113**, 045205 (2026).

3. **Evidence for the Collective Nature of Radial Flow in Pb+Pb Collisions with the ATLAS Detector** [*PRL*] [arXiv]
   ATLAS Collaboration. *Phys. Rev. Lett.* **136**, 032301 (2026).

4. **Impact of nuclear shape fluctuations in high-energy heavy ion collisions** [*EPJ A*] [arXiv]
   A. Dimri, S. Bhatta, and J. Jia. *Eur. Phys. J. A* **59**, 45 (2023).

Conference & Workshop Presentations

Measurements of radial and anisotropic flow in Pb+Pb collisions with the ATLAS detector 2026
Oral Presentation, Strangeness in Quark Matter (SQM 2026), Los Angeles, USA

Revealing the Bowling-Pin Geometry of Ne via O+O and Ne+Ne Collisions with ATLAS 2025
Oral Presentation, Initial Stages 2025, Taipei, Taiwan

Resolving the $C_4\{4\}$ Sign Change Puzzle in Pb–Pb Collisions 2025
Poster, Quark Matter 2025, Frankfurt, Germany

Effect of Nuclear Shape Parameters on Initial-State Observables in Heavy-Ion Collisions 2023
Poster, Quark Matter 2023, Houston, USA

Nuclear Shape Fluctuations in Heavy-Ion Collisions 2023
Talk, INT-23-1a: Intersection of Nuclear Structure and High-Energy Nuclear Collisions, Seattle, USA

Probing Effect of Nuclear Shape Coexistence in Heavy-Ion Collisions Using Glauber Model 2022
Oral Presentation, APS Division of Nuclear Physics Fall Meeting

Probing Effect of Nuclear Shape Coexistence in Heavy-Ion Collisions Using Glauber Model 2022
Poster, Strangeness in Quark Matter (SQM 2022)

Selected Summer Schools & Training Programs

Machine Learning for Fundamental Physics (ML4FP) Summer School, Georgia Institute of Technology 2026

ATLAS CAMPFIRE, Argonne National Laboratory 2024

National Nuclear Physics Summer School (NNPSS), University of California, Riverside 2023

National Nuclear Physics Summer School (NNPSS), Massachusetts Institute of Technology 2022

Honors, Awards & Fellowships

2026 **Andre-Mischke Award for Best Experimental Talk**, Strangeness in Quark Matter (SQM 2026).

2019 **Indian Academy of Sciences Summer Research Fellowship**.

2016–2019 **Academic Excellence Award**, School of Physical Sciences, NISER.

2017–2021 **Kishore Vaigyanik Protsahan Yojana (KVPY) Fellowship**, Department of Science and Technology, Government of India.

2016–2017 **INSPIRE Scholarship for Higher Education**, Department of Science and Technology, Government of India.

Experimental Collaboration & Service

ATLAS Collaboration, CERN 2022–2025

- Completed ATLAS authorship qualification through Run 3 heavy-ion electron and photon trigger-performance validation, including trigger-efficiency measurements and monitoring.
- Performed electron and photon trigger-performance studies for Pb+Pb and pp reference data, including tag-and-probe electron efficiencies and offline-photon validation.
- Studied the impact of disabling the Insertable B-Layer (IBL) on track reconstruction for O+O running, including validation of tracking efficiencies and fake rates using HIJING simulations.

STAR Collaboration, Brookhaven National Laboratory 2022–2025

- Served as Shift Leader during STAR data-taking operations, coordinating detector operations, data acquisition, and shift-crew responsibilities.
- Trained and supervised incoming Shift Leaders, evaluating their readiness for independent detector operations.

Teaching Experience

Graduate Teaching Assistant — General Chemistry Laboratory II (CHE 134) Spring 2022, Spring 2024

Stony Brook University

Led undergraduate laboratory sections, supervised experiments, guided data analysis and group projects, graded laboratory work, and held office hours.

Graduate Teaching Assistant — Molecular Science I (CHE 152) Fall 2021

Stony Brook University

Supported undergraduate workshops, held office hours, and graded homework assignments and examinations.

Technical Skills

Programming & Data Analysis: C++, Python, ROOT, Bash.

Scientific Computing: Linux, Git/GitHub, HTCondor/OSPool.

Simulation & Modeling: HIJING, TRENTo, Glauber Monte Carlo, MUSIC, iEBE-MUSIC, UrQMD.

ATLAS Software & Computing: Athena, Rucio, PanDA.